

Atividade 2: Auditoria Forense de Software e Plano de Resgate Técnico

Equipe: Dean Vinícius Palmeira de Meneses; Hilton Hunaldo Costa Silva Brito
Projeto: langfuse/langfuse

2026-05-31

1. Contexto e objetivo

Esta auditoria forense avalia a saúde gerencial e arquitetural do repositório langfuse/langfuse, com foco nos eixos do enunciado: Gestão (MPS.BR GPR), Anatomia do Código (SOLID/DRY) e Padrões GoF. Para cada item, apresentamos evidência, diagnóstico, classificação de risco e recomendação.

2. Eixo A: O Pulso da Gestão (MPS.BR GPR)

A1. Arqueologia de Issues

Evidência (issue e discussões): Issue #13933 e comentários com análise de causa e proposta de mudança de rota (novo endpoint by-id).

bug(sdk-python): get_dataset_runs() and related SDK methods fail with 400 when dataset name uses directory notation (slash) #13933

New issue

Open

#13952



simon376 opened 3 days ago

Describe the bug

Langfuse.get_dataset_runs(), Langfuse.get_dataset_run(), and any SDK method that embeds dataset_name as a URL path segment fail with 400 Bad Request (empty body) when the dataset name uses directory notation (e.g. regression/golden_v2).

Langfuse promotes slash-separated names as a feature for organizing datasets into directories. However, the SDK constructs URLs like:

```
GET /api/public/datasets/regression/golden_v2/runs
```

The server interprets golden_v2 as a sub-path rather than part of the dataset name. Percent-encoding the slash (regression%2Fgolden_v2) also fails with 400 — tested both via the SDK and via raw httpx requests directly against the API.

Expected behaviour: Dataset names using directory notation (containing /) should work correctly with all SDK methods that read run data.

Steps to reproduce

```
from langfuse import Langfuse

lf = Langfuse()

# Dataset named "regression/golden_v2" exists in Langfuse
# (created via the UI using directory notation)

runs = lf.get_dataset_runs(dataset_name="regression/golden_v2")
# → ApiError: status_code: 400, body: (empty)

# Same failure for:
# lf.get_dataset_run(dataset_name="regression/golden_v2", run_id="...")
# lf.api.datasets.get_runs(dataset_name="regression/golden_v2")
# lf.api.datasets.get_run(dataset_name="regression/golden_v2", run_id="...")

# Also fails when calling the REST API directly with httpx
# GET /api/public/datasets/regression%2Fgolden_v2/runs → 400
```

Note: these DO work (dataset_id as query param, not as path segment)
lf.api.datasets.list() → can resolve "2regression/golden_v2"
lf.api.dataset_run_items.list(dataset_id=resolved_id, run_id="...")
The gap is specifically the run-listing / run-reading endpoints

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Diagnostico: A issue mostra investigacao tecnica detalhada, confirmando falha tanto no SDK (encoding de path) quanto na rota do servidor, e culmina em mudanca de escopo para adicionar endpoints by-id. Essa dinamica indica boa engenharia investigativa, mas revela dependencia entre repositorios (SDK + server) e impacto em compatibilidade de API.

Risco: Medio. A dependencia cruzada e a mudanca de escopo podem atrasar correcoes e gerar regressao se nao houver alinhamento formal.

Recomendacao: Formalizar criterios de aceitacao multi-repo (SDK + server), vincular issues entre repositorios e registrar a decisao arquitetural (ADR). Incluir teste de integracao cobrindo nomes com “/” e a nova rota by-id.

A2. Gestao de Riscos Ocultos (TODO/FIXME)

Evidencia (TODOs relevantes):

otelIngestionQueue.ts L215-L219 (log de blob storage)

webhooks.ts L49-L52 (versionamento da API de webhook)

webhooks.ts L568-L570 (templates customizados via UI)

compileChatMessages.ts L98-L105 (IDs de mensagens)

langfuse / langfuse
Public
Notifications
Fork 2.9k
Star 28.3k

Code
Issues 328
Pull requests 339
Discussions
Actions

main
langfuse / worker / src / queues / webhooks.ts
Go to file

wochinge fix(worker): truncate oversized GitHub dispatch payloads (#1375...
b76b123 · 2 weeks ago

753 lines (671 loc) · 21.2 KB

```

1  import {
2    InternalServerError,
3    PromptWebhookOutboundSchema,
4    WebhookDefaultHeaders,
5    ActionExecutionStatus,
6    LangfuseNotFoundError,
7    JobConfigState,
8    isSlackActionConfig,
9    isWebhookAction,
10   isGitHubDispatchAction,
11   type AutomationDomain,
12   type ActionDomainWithSecrets,
13 } from "@langfuse/shared";
14 import { decrypt, createSignatureHeader } from "@langfuse/shared/encryption";
15 import { Prisma, prisma } from "@langfuse/shared/src/db";
16 import {
17   validateWebhookURL,
18   whitelistFromEnv,
19   fetchWithSecureRedirects,
20   WEBHOOK_URL_VALIDATION_LOG_CONTEXT,
21 } from "@langfuse/shared/src/server";
22 import {
23   TQueueJobTypes,
24   QueueName,
25   WebhookInput,
26   getAutomationById,
27   getActionByIdWithSecrets,
28
29
30
31
32
33 import { Processor, Job } from "bullmq";
34 import { backOff } from "exponential-backoff";
35 import { env } from "../env";
36 import { SlackMessageBuilder } from "../../features/slack/slackMessageBuilder";
37
38 // GitHub repository_dispatch client_payload: max 10 top-level properties and <64KB.
39 // https://docs.github.com/en/rest/repos/repos#create-a-repository-dispatch-event
40 const GITHUB_REPOSITORY_DISPATCH_MAX_PAYLOAD_BYTES = 64 * 1024;
41 const GITHUB_REPOSITORY_DISPATCH_TRUNCATION_MARKER =
42   "[TRUNCATED: GitHub repository_dispatch payload exceeded size limit]";
43 const GITHUB_REPOSITORY_DISPATCH_TRUNCATED_FIELDS = [
44   "prompt.prompt",
45   "prompt.config",
46 ];
47
48 // Handles both webhook and slack actions
49 export const webhookProcessor: Processor = {
50   job: Job<QueueJobTypes[QueueName.WebhookQueue]>,
51 } => {
52   try {
53     return await executeWebhook(job.data.payload);
54   } catch (error) {
55     logger.error("Error executing WebhookJob", error);
56     throw error;
57   }
58 };
59
60 // TODO: Webhook outgoing API versioning
61 export const executeWebhook = async (
62   input: WebhookInput,
63   options?: { skipValidation?: boolean },
64 ) => {
65   const { projectId, automationId } = input;
66
67   try {
68     logger.debug("Executing action for automation ${automationId}");
69
70     const automation = await getAutomationById(
71       projectId,
72       automationId,
73     );
74
75     if (!automation) {
76       logger.warn(
77         "Automation ${automationId} not found for project ${projectId}. We ack the job and will not retu
78       );
79       return;
80     }
81
82     // Route to appropriate handler based on action type
83     if (automation.action.type === "WEBHOOK") {
84       await executeWebhookAction({
85         input,
86         automation,
87         skipValidation: options?.skipValidation,
88       });
89     } else if (automation.action.type === "SLACK") {
90       await executeSlackAction({
91         input,
92         automation,
93       });
94     } else if (automation.action.type === "GITHUB_DISPATCH") {
95       await executeGitHubDispatchAction({
96         input,
97         automation,
98         skipValidation: options?.skipValidation,
99       });
100     } else {
101       throw new InternalServerError(
102         "Unsupported action type: ${automation.action.type}",
103       );
104     }
105
106     logger.debug(
107       "Action executed successfully for action ${automation.action.id}",
108     );
109   } catch (error) {
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111     throw error;
112   }
113 };
114
115 /**
116  * Shared HTTP execution logic for webhook and GitHub dispatch actions
117  * Handles: fetch with timeout, retry logic, status validation, execution tracking
118  */
119 export const executeHttpAction = (
120   url,
121   payload,
122   headers,
123   projectId,

```

langfuse / langfuse

Public

Notifications

Fork 2.9k

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<> Code

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webhooks.ts

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wochinge

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CodeBlameRawDownloadEditView

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117  * Handles: fetch with timeout, retry logic, status validation, execution tracking
118  */
119 export async function executeHttpAction({
120   url,
121   payload,
122   headers,
123   projectId,
```

Diagnostico: Existem TODOs em filas criticas (ingestao e webhooks) e em fluxo de LLM. Sao sinais de riscos tecnicos latentes que podem impactar confiabilidade e capacidade de auditoria (ex.: log de blob storage e versionamento de webhooks).

Risco: Medio/Alto. Os TODOs estao em areas de integracao com terceiros e pipelines de dados.

Recomendacao: Criar registro de riscos com priorizacao e dono, converter TODOs em issues com data-alvo e criterios de aceite, e definir politica de “no TODO without issue”.

A3. Ritmo de Entrega e Code Review

Evidencia (cadencia): Grafico de contribuicoes e Pulse.

Evidencia (code review): comentario de follow-up indicando ajustes por feedback em PR #13955.

Pulse
Contributors
Community standards
Commits
Code frequency
Dependency graph
Network
Forks

May 24, 2026 – May 31, 2026Period: 1 week

Overview

109 Active pull requests

26 Active issues

68 Merged pull requests

41 Open pull requests

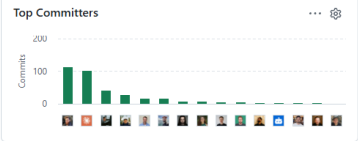
13 Closed issues

13 New issues

Summary

Excluding merges, **21 authors** have pushed **68 commits** to main and **237 commits** to all branches.

On main, **393 files** have changed and there have been **23,763 additions** and **9,253 deletions**



1 release published by 1 person

v3.176.0 v3.176.0
3 days ago

68 pull requests merged by 17 people

- fix(search): match escaped unicode content in full-text search - Issue #11538**
#13644 merged 2 days ago
- fix: refine LANGFUSE_DISABLE_LEGACY_TRACING_IO_SEARCH handling**
#13929 merged 2 days ago
- perf(eval): skip FINAL and unused aggregations in checkTraceExistsAndGetTimestamp**
#13934 merged 2 days ago
- perf(blob-export): replace observations FINAL with LIMIT 1 BY**
#13935 merged 2 days ago
- feat(ui): notification for MCP v2**
#13895 merged 3 days ago
- fix(eval): stop retrying context overflow errors**
#13930 merged 3 days ago
- fix(mcp): Do not use intersection or union JSON schemas**
#13927 merged 3 days ago
- feat: introducing LANGFUSE_DISABLE_LEGACY_TRACING_IO_SEARCH flag to allow disabling FTS on v3 tables**
#13912 merged 3 days ago
- fix(mcp): Missing audit logs**
#13915 merged 3 days ago
- fix: tighten matches operator semantics**
#13928 merged 3 days ago
- fix(api,mcp): Stabilize score config pagination order**
#13632 merged 3 days ago
- fix(mcp): Use ids instead of names for dataset tools**
#13916 merged 3 days ago
- ci(deps): bump the github-actions group with 3 updates**
#13924 merged 3 days ago
- fix(mcp): Throw if not exists in 'deleteAnnotationQueueAssignment'**
#13914 merged 3 days ago
- fix(mcp): Rename relevant tools from create to upsert**
#13913 merged 3 days ago
- feat(model-prices): add claude-opus-4-8**
#13919 merged 3 days ago
- fix(traces): Infinite recursion in 'buildStepGroups'**
#13900 merged 3 days ago
- chore(mcp): Update MCP server version**
#13883 merged 3 days ago
- feat(ui): notification for code evals launch**
#13894 merged 3 days ago
- ci: notify slack on critical workflow failures**
#13910 merged 3 days ago
- chore(deps): dedupe and bump**
#13911 merged 3 days ago
- ci(sdk): use repo token for SDK spec workflow**
#13909 merged 3 days ago
- refactor(mcp,api): Create shared services**
#13879 merged 4 days ago
- fix(llm): harden LLM connection fetches**
#13797 merged 4 days ago
- fix(evals): tighten eval log IO cell padding**
#13906 merged 4 days ago
- fix(dashboards): drop scoreName filter on traces and observations views**
#13901 merged 4 days ago
- chore: Add .superset/ to .gitignore**
#13905 merged 4 days ago
- docs: Update test running instructions**
#13902 merged 4 days ago
- fix(evals): improve code evaluator formatting**
#13897 merged 4 days ago
- fix(prompts): don't comment fetching on many prompt versions**

fix(tracing): treat events-backed projects as tracing configured #13955

 Open

zamal-db wants to merge 2 commits into

langfuse:main from zamal-db:fix/hasanytrace-events-fallback



 Conversation 5

 Commits 2

 Checks 0

 Files changed 2

+60 -16 



zamal-db commented yesterday • edited • ...

What this PR does

Improves tracing setup-state detection for events-backed ingestion.

Problem I hit

While running agent-oss style workflows on Langfuse, I hit a setup mismatch: tracing data was already present via events-backed ingestion, but setup was still showing "tracing not configured".

Root cause: `hasTracingConfigured` relies on `hasAnyTrace(projectId)`, and `hasAnyTrace` was only checking ClickHouse `traces`.

How I fixed it

I updated `hasAnyTrace` to resolve tracing presence in this order:

1. PostgreSQL fast path: `project.hasTraces`
2. ClickHouse `traces`
3. ClickHouse events fallback: `hasAnyEvent(projectId)`

If either traces or events is present, it now persists `project.hasTraces = true`.

Why this matters

This removes a real false-negative setup state:

- tracing data exists,
- setup no longer says "not configured".

Type of change

- ☒ Bug fix (non-breaking change which fixes an issue)
- ☒ Refactor (deduplicated persistence block, no behavior change beyond fix)

Validation

Focused suite:

9

- `src/__tests__/server/repositories/hasAnyTrace.serverTest.ts`
 - PG fast path

Reviewers

 claude[bot]

+1 more reviewer

 greptile-apps[bot]

Assignees

No one assigned

Labels

bug-fix-mar-26

size:S

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

2 participants

Diagnostico: A atividade do repositório é contínua e há evidências de feedback e iteração em PRs. Entretanto, vários PRs recentes mostram apenas comentários de bots, o que pode indicar revisão humana inconsistente.

Risco: Médio. A cadência é boa, mas a ausência de revisão humana explícita em parte dos PRs aumenta o risco de regressão.

Recomendação: Reforçar política de code review (mínimo 1 aprovação humana), checklist de revisão e evidências de testes executados em PR.

3. Eixo B: Anatomia do Código (SOLID/DRY)

B1. Teste da Troca (DIP)

Evidência (pontos de acoplamento):

LLMAdapter enum

fetchLLMCompletion.ts L303-L413 (if/else por adapter)

llm-api-key router L100-L140 (branch por adapter)

useModelParams.ts L220-L269 (switch por adapter)


[langfuse / langfuse](#)
Public

🔔 Notifications
🍴 Fork 2.9k
★ Star 28.3k

<> Code
🔍 Issues 328
🔗 Pull requests 339
💬 Discussions
🔗 Actions
⋮

📁 main
langfuse / packages / shared / src / server / llm
🔍 Go to file
⋮

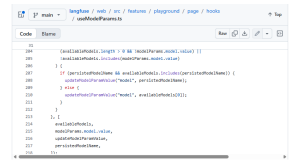
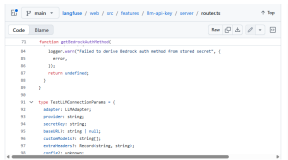
👤 hasiebpt
feat(model-prices): add claude-opus-4-8 (#13919)
✓
ca84928 · 3 days ago
🕒

579 lines (532 loc) · 16.6 KB

```

1  import { LlmApiKeys } from "@prisma/client";
2  import z from "zod";
3  import {
4    BedrockConfigsSchema,
5    VertexAIConfigsSchema,
6  } from "../interfaces/customLLMProviderConfigsSchemas";
7  import { JSONObjectSchema } from "../utils/zod";
8  import type {
9    InternalTraceEventInput,
10   InternalTraceExperimentContext,
11 } from "../internalTraceEvents";
12
13 // disable lint as this is exported and used in web/worker
14
15 export const LLMJSONSchema = z.record(z.string(), z.any());
16 export type LLMJSONSchema = z.infer<typeof LLMJSONSchema>;
17
18 export const JSONSchemaForSchema = z
19   .string()
20   .transform((value, ctx) => {
21     try {
22       const parsed = JSON.parse(value);
23       return parsed;
24     } catch {
25       ctx.addIssue({
26         code: "custom",
27         message: "Parameters must be valid JSON",
28       });
29       return z.NEVER;
30     }
31   })
32   .pipe(
33     z
34       .object({
35         type: z.literal("object"),
36         properties: z.record(z.string(), z.any()),
37         required: z.array(z.string()).optional(),
38         additionalProperties: z.boolean().optional(),
39       })
40       .loose()
41       .transform((data) => JSON.stringify(data, null, 2)),
42   );
43
44 export const LLMToolDefinitionSchema = z.object({
45   name: z.string(),
46   description: z.string(),
47   parameters: LLMJSONSchema,
48 });
49 export type LLMToolDefinition = z.infer<typeof LLMToolDefinitionSchema>;
50
51 export const LLMToolCallSchema = z.object({
52   name: z.string(),
53   id: z.string(),
54   args: z
55     .record(z.string(), z.unknown())
56     .nullable()
57     .transform((val) => val ?? {}),
58 });
59 export type LLMToolCall = z.infer<typeof LLMToolCallSchema>;
60
61 export const OpenAIToolCallSchema = z.object({
62   id: z.string(),
63   function: z.object({
64     name: z.string(),
65     arguments: z.union([
66       z.record(z.string(), z.unknown()),
67       z
68         .string()
69         .transform((v) => {
70           try {
71             return JSON.parse(v);
72           } catch {
73             return v;
74           }
75         }),
76     ]),
77   }),
78 });

```



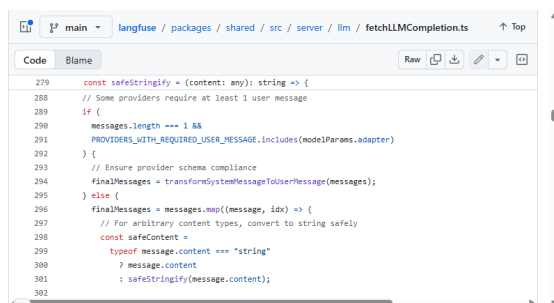
Diagnostico: A troca de provedor de IA impacta diversas camadas (shared, server e UI). A logica especifica de cada adapter esta espalhada, o que viola o Principio de Inversao de Dependencia e aumenta o custo de mudanca.

Risco: Alto. A troca de provedor exige alteracoes em varios arquivos e aumenta risco de inconsistencias.

Recomendacao: Criar um registry de provedores (Strategy/Factory), centralizando configuracao e regras por adapter; expor uma interface unica para UI consumir, reduzindo branches.

B2. Busca por “God Objects”

Evidencia (orquestracao centralizada): fetchLLMCompletion.ts L303-L413.



Diagnostico: O arquivo concentra configuracao de proxy, timeouts, retries e instanciacao de diferentes clientes por provider. Isso indica uma unidade com multiplas responsabilidades.

Risco: Medio/Alto. Mudancas para um provider podem afetar outros por acoplamento no mesmo fluxo.

Recomendacao: Extrair classes por provider (ex.: AnthropicStrategy, OpenAIStrategy) e deixar o orquestrador apenas selecionar e delegar.

B3. DRY (duplicacao)

Evidencia (duplicacao declarada): tree-building.ts L466 (duplicado em TraceTree e TraceTimeline).

```
/**
 * Tree building utilities for trace component.
 *
 * IMPLEMENTATION APPROACH:
 * - Uses fully iterative algorithms (no recursion) to avoid stack overflow on deep trees (10k+ depth).
 *
 * Algorithm Overview:
 * 1. Sort observations by startTime
 * 2. Build dependency graph: Map-based parent-child relationships (O(N))
 * 3. Topological sort: Process nodes bottom-up (leaves first) using queue with index-based traversal
 * 4. Cost aggregation: Compute bottom-up during tree construction (children before parents)
 * 5. Flatten to searchItems: Iterative pre-order traversal using explicit stack
 *
 * Level filtering (hiding DEBUG observations etc.) is NOT done here - it's applied
 * as a post-processing step via removeHiddenNodes() in the display layer (TraceDataContext).
 *
 * Complexity: O(N) time, O(N) space - handles unlimited depth without stack overflow.
 *
 * Main export: buildTraceUIData() - builds tree, nodeMap, and searchItems from trace + observations.
 */

import { type TreeNode, type TraceSearchListItem } from "../types";
import { type ObservationReturnType } from "@src/server/api/routers/traces";
import Decimal from "decimal.js";
import {
  type ObservationLevelType,
  ObservationLevel,
} from "TraceTimeline";
```

Diagnostico: O proprio TODO indica duplicacao de logica. Essa repeticao tende a gerar divergencia de comportamento e custos de manutencao.

Risco: Medio.

Recomendacao: Extrair utilitario compartilhado e adicionar testes unitarios para garantir equivalencia.

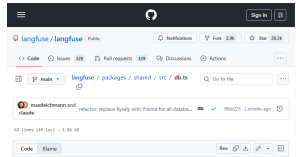
4. Eixo C: Padroes de Projeto (GoF)

C1. Criacionais

Evidencia (Singleton):

PrismaClientSingleton

SlackService.getInstance



Diagnostico: O uso de Singleton garante instancia unica de recursos caros (DB e Slack). Embora seja adequado em alguns pontos, adiciona estado global e pode dificultar testes.

Risco: Baixo/Medio.

Recomendacao: Manter Singleton onde houver justificativa de custo, mas permitir injecao de dependencias em testes e documentar ciclo de vida.

C2. Estruturais

Evidencia (Adapter): chatml/adapters/index.ts L11-L37.

```
1 import type { NormalizerContext, ProviderAdapter } from "../types";
2 import { langgraphAdapter } from "../langgraph";
3 import { aisdskAdapter } from "../aisdk";
4 import { openAIAdapter } from "../openai";
5 import { geminiAdapter } from "../gemini";
6 import { microsoftAgentAdapter } from "../microsoft-agent";
7 import { semanticKernelAdapter } from "../semantic-kernel";
8 import { pydanticAIAdapter } from "../pydantic-ai";
9 import { genericAdapter } from "../generic";
```

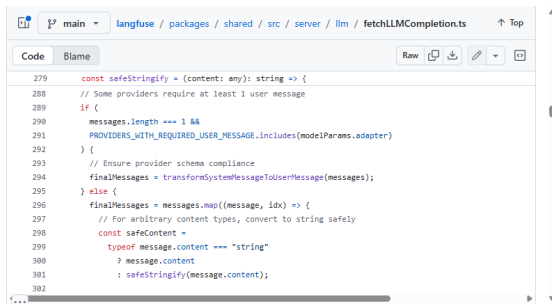
Diagnostico: O codigo organiza adapters para normalizar diferentes frameworks. Isso atende bem ao padrao Adapter, mas requer governanca (ordem, deteccao e fallback).

Risco: Baixo.

Recomendacao: Formalizar interface dos adapters, documentar criterios de deteccao e adicionar testes de regressao.

C3. Comportamentais

Evidencia (cadeia de if/else por provider): fetchLLMCompletion.ts L314-L413.



Diagnostico: A ausencia de um Chain of Responsibility/Strategy centralizado exige alteracao no core para qualquer novo provider, o que cresce linearmente com o numero de adapters.

Risco: Medio.

Recomendacao: Aplicar Strategy + Registry (ver plano de resgate) para substituir o encadeamento de condicoes.

5. Plano de Resgate Sugerido

5.1 Refatoracao Conceitual (GoF)

Trecho alvo: `fetchLLMCompletion.ts`. O objetivo e isolar detalhes por provider e reduzir acoplamento. Exemplo conceitual:

```
interface LLMProviderStrategy {
  id: LLMAdapter
  buildClient(ctx): ChatModel
  normalizeResponse(res): NormalizedResponse
}
```

```
const registry = new Map<LLMAdapter, LLMProviderStrategy>([
  [LLMAdapter.OpenAI, new OpenAISTrategy()],
  [LLMAdapter.Anthropic, new AnthropicStrategy()],
  ...
])
```



```
export async function fetchLLMCompletion(params) {
  const strategy = registry.get(params.modelParams.adapter)
  if (!strategy) throw new Error("Unsupported adapter")
  const client = strategy.buildClient(params)
  return client.call(...)
}
```

Benefícios: reduz alterações no core, melhora testabilidade e permite adicionar novos providers sem editar a lógica central.

5.2 Roadmap de Maturidade (MPS.BR Nivel G)

- Acao 1: GPR - Planejamento e acompanhamento:** criar baseline de planejamento (roadmap trimestral, milestones, criterios de aceite) e registrar progresso em GitHub Projects com responsaveis e metas.
- Acao 2: GRE - Rastreabilidade de requisitos:** mapear requisitos da A1 para issues/PRs, padronizar templates e adicionar links cruzados (issue → PR → teste).
- Acao 3: Gestao de riscos e qualidade:** manter registro de riscos (TODO/FIXME viram issues), definir politica de code review (minimo 1 aprovacao humana) e checklist de testes obrigatorios.